Tolga Arslan

Full-Stack Developer · React · Node.js · Express · Python · FastAPI
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Profile Summary

Computer Engineering graduate (English-taught) building full-stack web applications. I develop React frontends and Node.js/Express or FastAPI backends, secure them with cookie-based sessions, rate limiting and validation, cover them with automated tests and ship them through GitHub Actions CI to Vercel and Render. Interned in Agile teams at ArVis (Teknopark İstanbul) and VBT Yazılım.

Technical Skills

Languages: JavaScript, Python, SQL, C# (basic)
Frontend: React 18, Vite, React Router, Axios, HTML, CSS (responsive, accessible), Streamlit
Backend: Node.js, Express.js, FastAPI, REST API design, JWT / bcrypt, Mongoose
Databases: MongoDB (Atlas), PostgreSQL (SQLAlchemy, Alembic), SQL Server
Testing / DevOps: Jest, Supertest, Vitest, React Testing Library, pytest, ESLint, GitHub Actions, Docker, Vercel, Render
Data / ML: Pandas, scikit-learn, LightGBM, SHAP
Fundamentals: Object-Oriented Programming, Data Structures and Algorithms, Software Engineering, Agile
Languages (Spoken): Turkish — Native; English — B2 (Upper-Intermediate), English-taught B.Sc. program

Projects

Pawfect Care — Pet Care Platform | GitHub | Live Demo Feb 2025 – Oct 2026

React, Vite, Node.js, Express.js, MongoDB, Jest, Vitest, GitHub Actions — capstone project, extended in 2026

- Built a full-stack app for tracking pets' health records, vaccines (with overdue and upcoming booster alerts), weight trends, medications and vet appointments, plus a community blog; the React frontend has lazy-loaded routes, light/dark themes and accessible forms.
- Designed a REST API with 34 endpoints secured by httpOnly SameSite=Strict JWT cookies, session revocation, brute-force rate limiting, helmet/CSP headers and server-side validation; added account data export and deletion with a KVKK privacy notice.
- Wrote ~300 tests (170+ Jest/Supertest API tests, 120+ Vitest/Testing Library UI tests) run with ESLint on Node 22 and 24 in GitHub Actions; deployed the frontend on Vercel and the API on Render with MongoDB Atlas.

HavaUyarı — Air Quality Early Warning System | GitHub Sept 2026 – Oct 2026

Python, FastAPI, PostgreSQL, Docker, Streamlit, pytest, LightGBM

- Built a Python system that ingests data from the Ministry's SİM stations and three Open-Meteo APIs and serves 24-hour PM2.5 forecasts and early warnings for 8 stations in 5 cities through a FastAPI service with an hourly cache and a Streamlit dashboard.
- Added PostgreSQL storage for forecasts and alert history (SQLAlchemy, Alembic) and docker-compose; 144 tests, incl. PostgreSQL integration tests, run in CI; backtest MAE is 47.7% below raw CAMS.

Card Fraud Detection | GitHub | Live Demo Sept 2026 – Oct 2026

Python, FastAPI, Streamlit, Docker, LightGBM, pytest

- Built a FastAPI scoring service and a Streamlit panel for a LightGBM model trained on 1.85M synthetic transactions (Kaggle Sparkov), returning a decision and the top 3 reasons per transaction; a cost-based alert rule caught 98% of fraud value.
- Containerized the app with Docker and fixed per-session memory growth (+205 MB over 3 sessions) found with docker stats under a 2 CPU / 3 GB limit; published a public live demo.

Experience

DeepTech/AI Intern, ArVis Technology – Teknopark İstanbul July 2024 – Aug 2024

In collaboration with Roket Technology & Cubeforward

- Prepared training data for the CNN-based model of "Neurovisionist", an AI-powered early Alzheimer's diagnosis project, within a 16-person R&D team; performed data cleaning and preprocessing to improve model accuracy.
- Developed cognitive-exercise mini apps targeting memory functions; worked in 6 weeks of Agile sprints.

Software Intern, VBT Yazılım A.Ş. July 2023 – Aug 2023

- Trained in web UI (HTML, CSS, JavaScript, Angular) and mobile (Flutter) development; learned ITSM/ITIL processes.
- Completed hands-on exercises with C#, .NET Core and SQL Server, managing version control with Git.

Military Service

Compulsory Military Service — Completed Dec 2025 – May 2026

Education

Bandırma Onyedi Eylül University, B.Sc. in Computer Engineering (English) Sept 2020 – Feb 2025

Certifications

ArVis DeepTech/AI 2024 Internship — Certificate of Achievement • İş Bankası ProSchool Data & AI Class (Toptalent.co) • SQL Server Database Programming (Udemy) • Flutter App Development (Google Digital Garage)